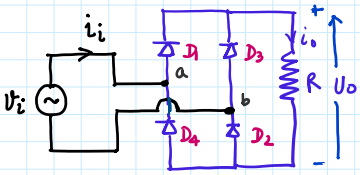
Full Wave Rectifier:



Full bridge rectifier

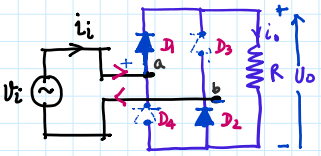
During positive half cycle D_1-D_2 are forward biased and hence conduct, with negligible source impedance

$V_o \simeq V_i$ ——— (*)

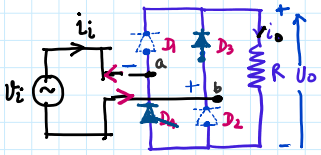
D_3-D_4 are reversed biased.

During negative half cycle

V_b is positive and V_a is negative. Thus D_3- D_4 are forward biased with $V_o = -V_i$.



(a) Positive half cycle

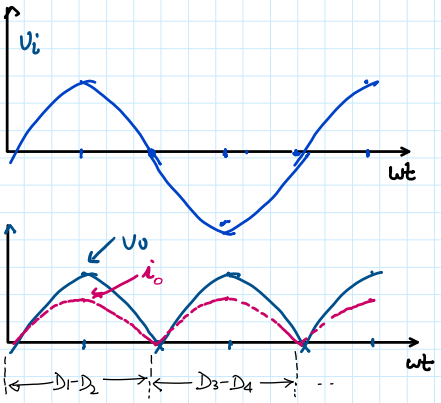


(b) Negative half Cycle

D_1-D_2 are reversed biased.

During non-conduction: Non-conducting diodes are subjected to maximum reverse bias voltage of V_m. This is known as Peak Inverse Voltage.

PIV = $-V_m$



For this rectifier circuit:

$$V_o = V_{avg} = \frac{1}{T}\int_0^T V_o\,dt = \frac{2 \cdot V_m}{\pi}$$

$$V_{rms} = \left[\frac{1}{T}\int_0^T V_o^2\,dt\right]^{1/2} = \frac{V_m}{\sqrt{2}}$$

$$FF = \frac{V_{orms}}{V_{oavg}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = 1.11$$

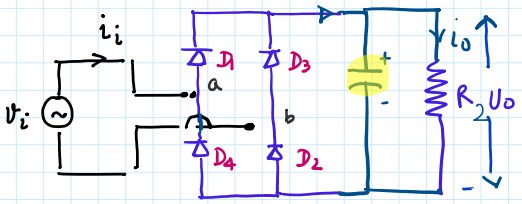
$$RF = \sqrt{FF^2 - 1} = \sqrt{0.2337} = 0.4834$$

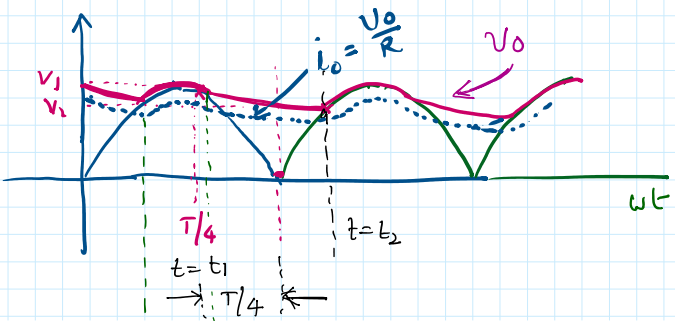
Current i_o is similar to V_o for resistive load.

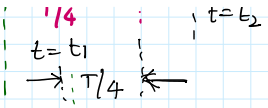
$$i_o = \frac{V_o}{R}$$

$$I_o = \frac{2\,I_m}{\pi}, \quad I_{rms} = \frac{I_m}{\sqrt{2}}$$

Full Wave Rectifier with Capacitance:







$V_2 = V_1 e^{-t/R_2C}$

$V_1 \simeq V_{max}$

$t \simeq T/4$

$V_2 = V_m * e^{-\frac{0.005}{0.1}}$

$= V_m e^{-0.05} = 0.95 V_m$

$V_r = (V_1 - V_2) = V_m - 0.95 V_m = 0.05 V_m$

$V_{r,max} = \frac{V_r}{2} = \frac{V_1 - V_2}{2} = \frac{V_m - 0.95 V_m}{2} = 0.025 V_m$

$V_{r,rms} = \frac{V_{r,max}}{\sqrt{2}} = \frac{0.025 V_m}{\sqrt{2}} = 0.0176 V_m$

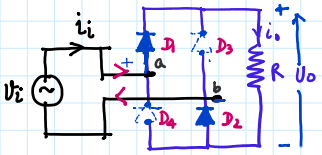
$V_{oav} = \frac{V_1 + V_2}{2} = \frac{V_m + 0.95 V_m}{2} = 0.975 V_m$

$\therefore RF = \frac{V_{r,rms}}{V_{oav}} = \frac{0.0176 V_m}{0.975 V_m} = 0.018$

		Ripple, V_r	Ripple factor (RF)
Half wave Rectifier	without C	V_m	1.21
	with C	$0.14 V_m$	0.05
Full wave Rectifier	without C	V_m	0.48
	with C	$0.05 V_m$	0.018

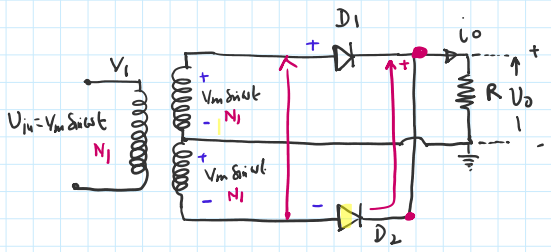
Peak Inverse Voltage: Peak inverse voltage (PIV) is defined as the maximum voltage, when diode is reverse biased (i.e., not conducting).

PIV for full wave rectifier = V_m



(a) Positive half cycle

Another Circuit for Full Wave Rectifier:

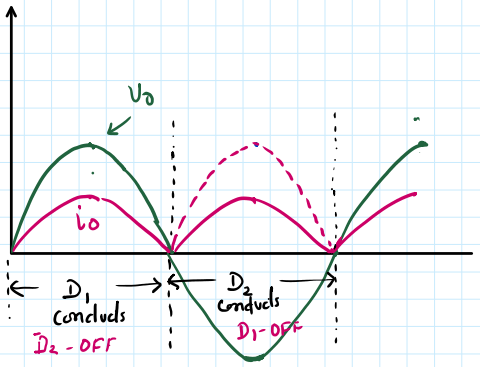


Center-tapped T/F

PIV = Peak Inverse Voltage

Center-tapped T/F

PIV = Peak Inverse Voltage

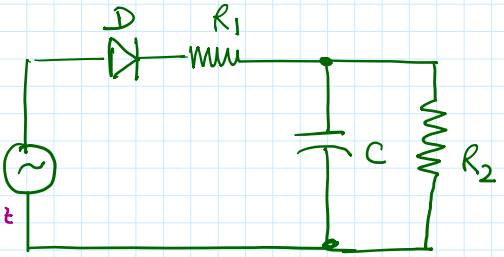


PIV = $2V_m$

Hence reverse bias rating of diode will be more.

Example : C-filter Design.

Consider a half wave rectifier. It is desired that the ripple at the output should be 10% of the peak value of the applied voltage. What is the value of capacitor for resistive load $R_2 = 1\ k\Omega$. Consider R_1 very small value.



$$V_0 = V_m e^{-(t-T/4)/R_2C}$$

$$V_2 = V_m e^{-(T-T/4)/R_2C}$$

$$(V_1 - V_2) = 0.1\,V_m = V_m - V_m e^{-(3T/4)/R_2C}$$

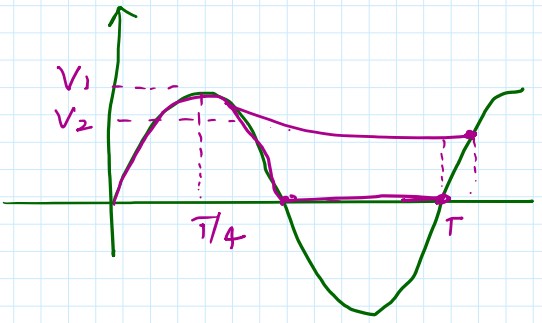
$$0.1 = 1 - e^{-3 \times \frac{0.02}{4} \times \frac{1}{(10^3 \times C)}}$$

$$0.9 = e^{-15 \times 10^{-6}/C}$$

$$\ln 0.9 = -\frac{15 \times 10^{-6}}{C}$$

$$\Rightarrow -\frac{15 \times 10^{-6}}{C} = -0.1$$

$$\boxed{C = 150\ \mu F}$$



Example 2 : Now consider full wave rectifier with 5% ripple in the output voltage to that of the

5% ripple in the output voltage to that of the peak supply voltage. Design the filter C.

For full wave rectifier,

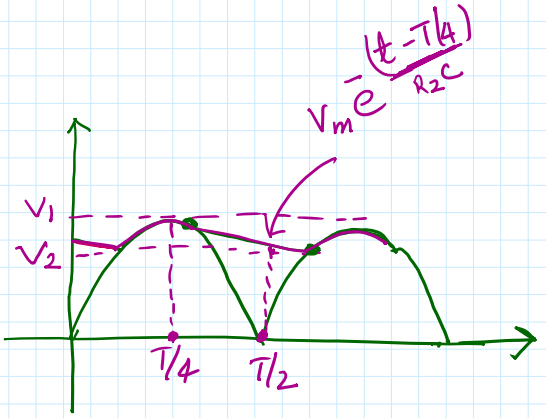
$$V_2 = V_m e^{-(t-T/4)/R_2C}$$

$$= V_m e^{-(T/2-T/4)/R_2C}$$

$$= V_m e^{-T/(4R_2C)}$$

$$= V_m e^{-\frac{0.02}{4\times1\times10^{-3}C}}$$

$$V_2 = V_m e^{-5\times10^{-6}/C}$$



$$\text{Ripple} = V_r = V_1 - V_2 = 0.05V_m = V_m - V_m e^{-5\times10^{-6}/C}$$

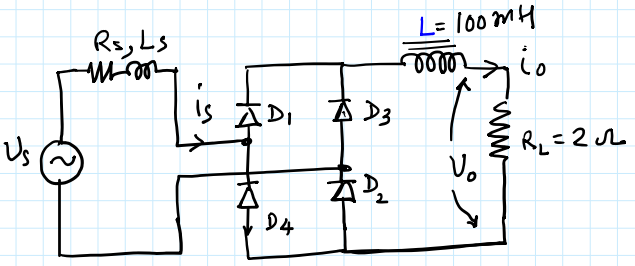
$$\Rightarrow \quad 0.95 = e^{-5\times10^{-6}/C}$$

$$-\frac{5\times10^{-6}}{C} = \ln(0.95) = -0.0513$$

$$\Rightarrow \quad C = 97.47\ \mu F$$

$$\boxed{C \simeq 100\ \mu F}$$

Example: Consider the following rectifier circuit which supplies constant current to the resistive load.



The circuit supplies constant current to load Resistance R.
To make current constant L is taken large.
Draw output voltage and current and the input voltage and current.

$V_s = 230\sqrt{2}\sin 314t$

Ideal source: $R_s \to 0$, $L_s \to 0$

$I_o = \frac{V_o}{R}$; V_o = average value of v_o

v_i i_i

$I_o = \frac{V_o}{R}$; V_o = average value of V_o

$= \frac{2V_m}{\pi} \times \frac{1}{R}$

$= \frac{2 \times 230\sqrt{2}}{3.14} \times \frac{1}{2}$

$= 103.5\,A$

$I_{orms} = I = 103.5\,A$

